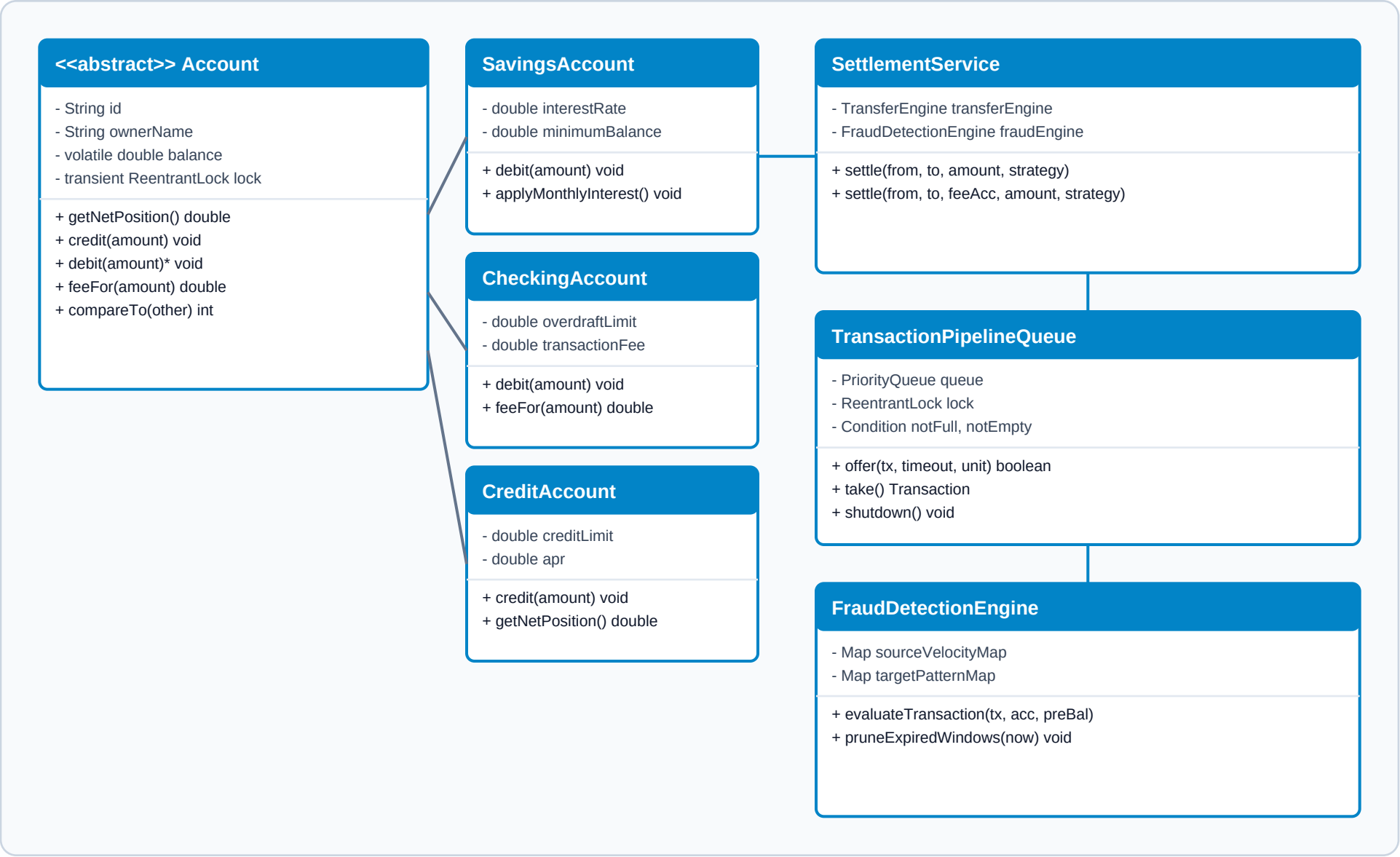


Evolved System Architecture Class Diagram (UML v2.0)

CS-212 Final Project: Automated Banking & Real-Time Fraud Engine | Author: Muhammad Arslan



Key Architectural Refactorings & Specifications (v2.0)

Component	Refactored Behavior & Pattern	Defect Addressed
Account / CreditAccount	credit() overridden to reduce debt; getNetPosition() returns signed liquidity contribution	F3 (BLOCKER)
SettlementService	Pre-execution fraud evaluation on pre-transfer balance before settlement	F1 (BLOCKER)
FraudDetectionEngine	Rapid drain threshold fixed to 90% pre-balance; atomic Deque monitor synchronization	F2, F14 (MAJOR)
PipelineCoordinator	Active bounded queue (100) with 3 consumer worker threads and FX event publishing	D1 (BLOCKER)
AccountLockManager	Deterministic total ordering on N accounts by ID sorting (compareTo)	F10, Task 2.3